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SECOND EDITION

ENTERPRISE SYSTEMS FOR MANAGEMENT

CHAPTER 2

SYSTEMS INTEGRATION

Learning Objectives

- Understand the impact of organizational structure on information systems.
- Find out about the types of functional silos in organizations.
- Learn about the evolution of information systems technology generations and architectures and its influence on silo environment.
- Know what systems integration is and why it is important for organizations.
- Understand the role of Enterprise Resource Planning (ERP) systems in systems integration.

Preview

- Systems integration means that you allow a heterogeneous (hodgepodge) IS to communicate or integrate and share information (or data) **seamlessly** with one another.
- Systems integration is a key issue for an organization for its growth
 - Management needs to pay close attention to this issue.
- ERP systems are a major kind of enterprise information system allowing organizations to integrate different systems into one organization-wide application with an integrated database management system.



Source: Pearson Education, Inc. Publishing as Prentice Hall

Functional Silos

- Silos are basically compartmentalized operating units isolated from their environment.

Horizontal Silos

- The POSDCORB (Planning, Organizing, Staffing, Directing, Coordinating, Reporting and Budgeting) categorization by Luther Gulick led to a set of formal organization functions such as control, management, supervision, and administration starting in late 1930s.
- Classification of organizations into departments like Accounting and Human Resources, reflects the breaking of complex tasks into smaller manageable tasks that could be assigned to a group of people who could then be held responsible.

Figure 2-1 Functional Model of Organization (POSDCORB)

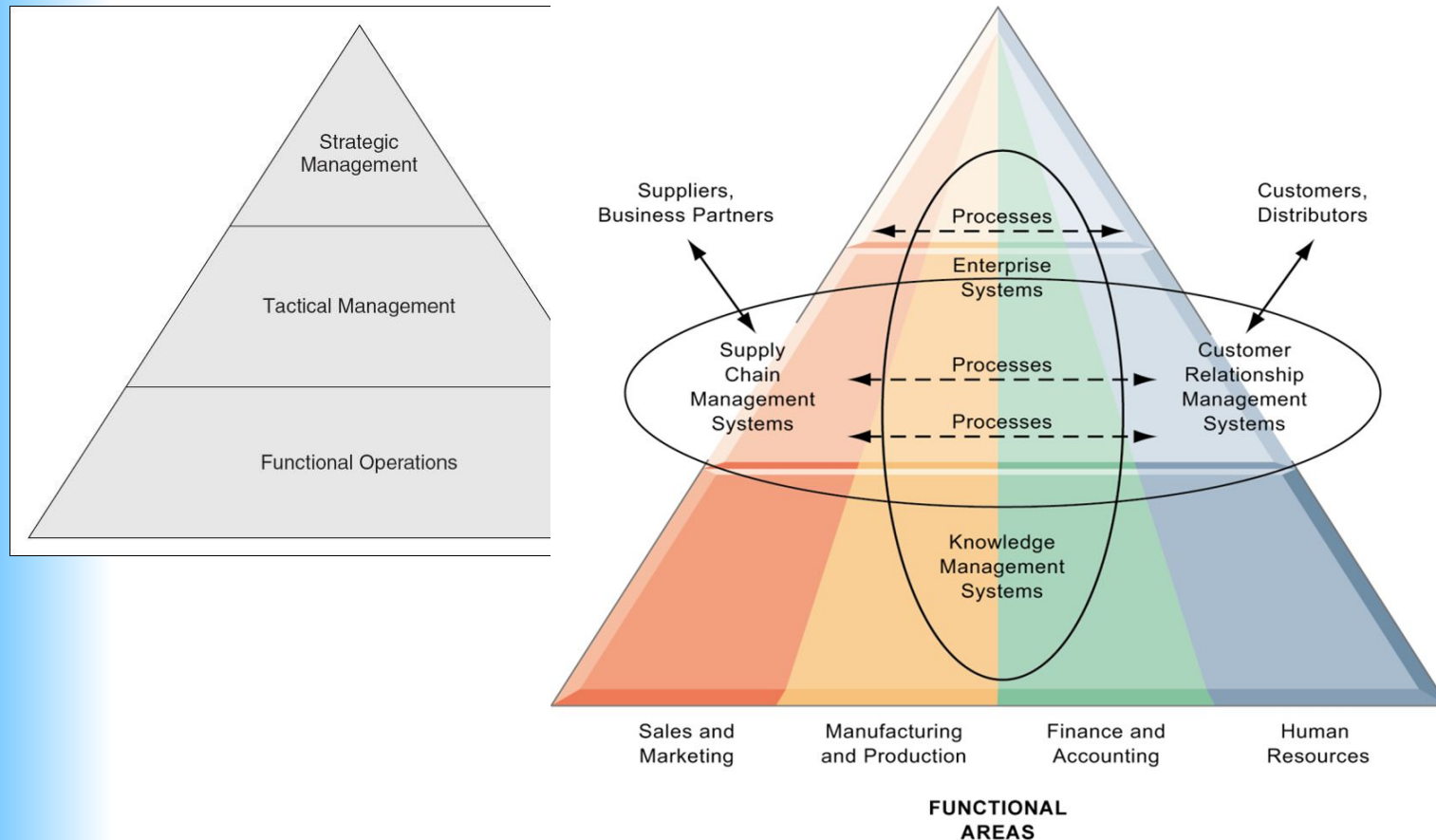


Functional Silos (Cont'd)

Vertical Silos

- Organizations also divided roles in hierarchical layers from strategic planning to management control and operation control.
- CEOs and Presidents plan long-term strategy, midlevel management focuses on tactical issues and on the execution of organizational policy whereas the lower-level management task is to focus on the day-to-day operations of the company.

Figure 2-2 Hierarchical Model of Organization

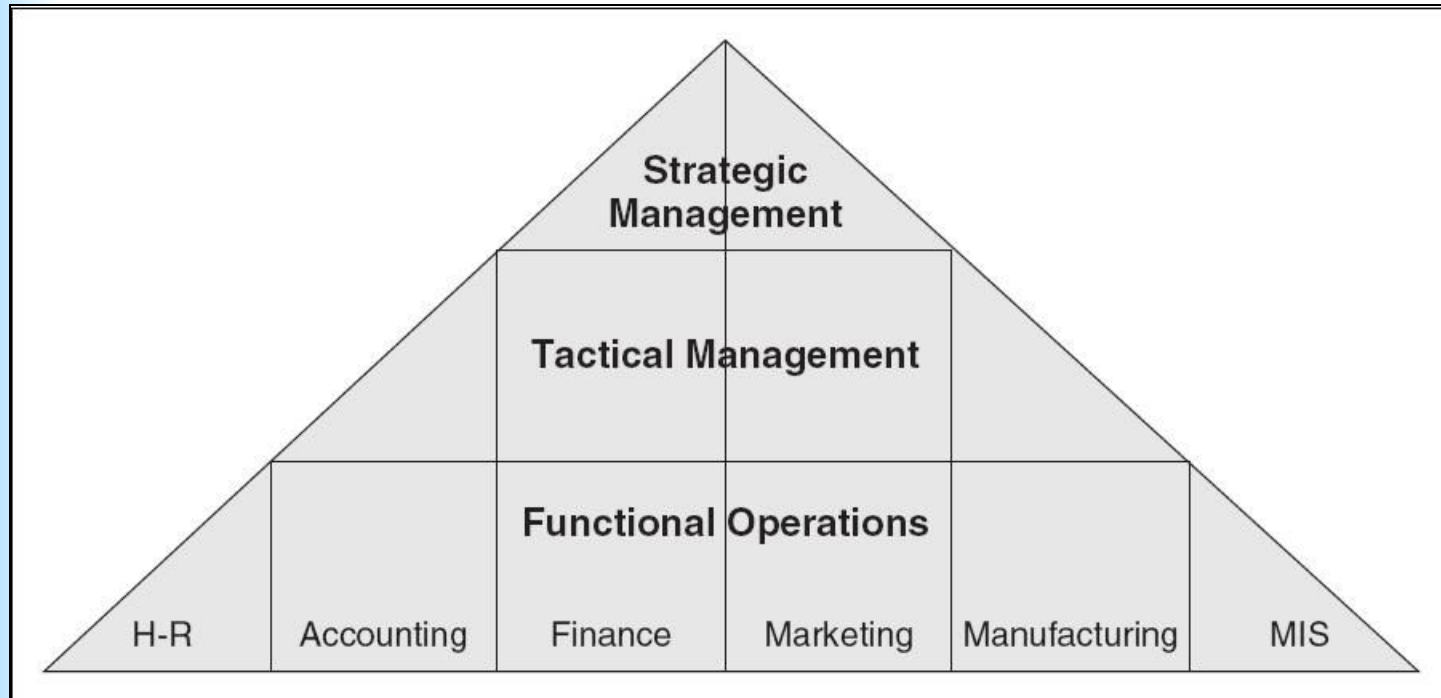


Business Process and Silos

- The problem of functional silos gave birth to business process re-engineering (BPR).
- The cross-functional business process can involve people and resources from various functional departments working together, sharing information at any level of the organization.
- The cross-functional organizational structure breaks the functional silos by opening up the informational flows from one department to another.



Figure 2-3 Matrix Structure of Organization



Evolution of Information Systems in Organizations

- The functions of an organization (e.g., sales, manufacturing, and HR) are important as they provide a structure by which an organization functions smoothly.
- A silo information system is inefficient, inaccurate, and expensive.
 - The system creates bottlenecks for everyone and information is not available in real-time.
- The evolution of IS suggests that its role has generally been to support evolving information needs of the organization.

THE EVOLUTION OF INFORMATION SYSTEMS



1970's

Database management systems
Decision support
Office information systems



1990's

Enterprise information systems
E-Commerce systems
Knowledge Management systems
BI systems (EIS)

1950's

Transaction processing
Reporting systems



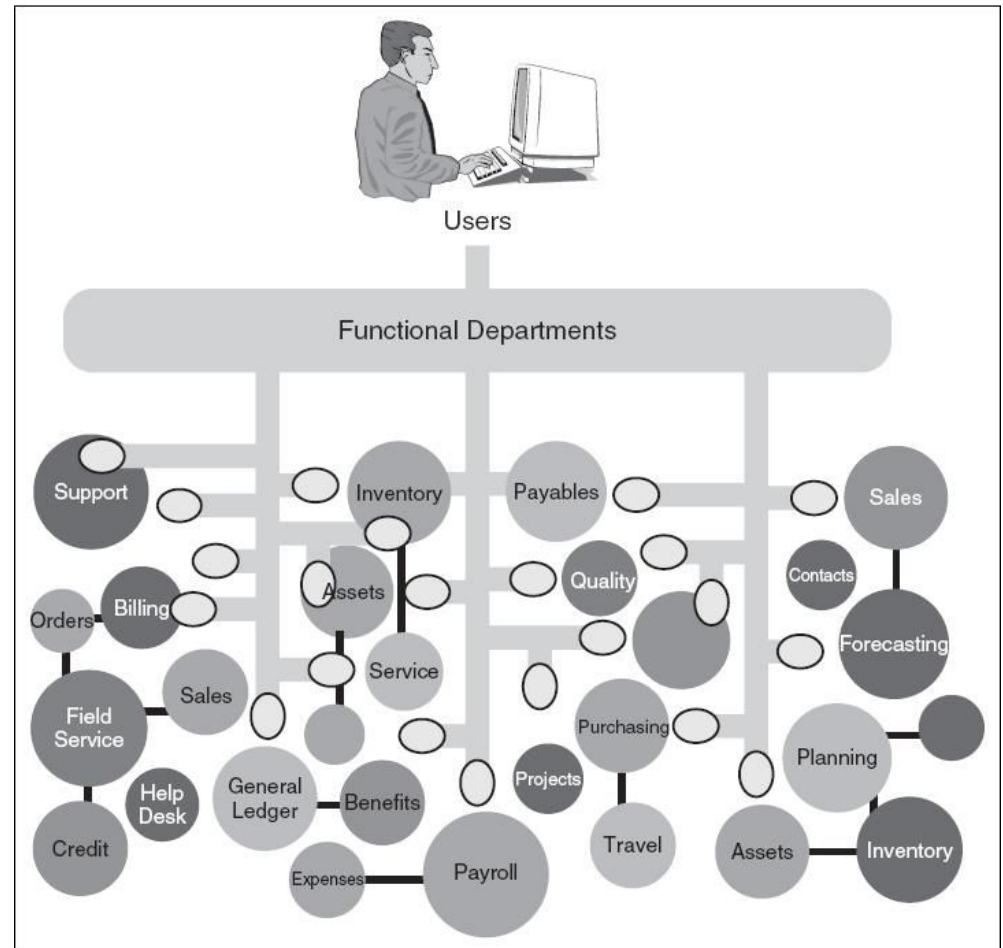
1980's

Database management systems
AI based systems
Executive information systems
Strategic systems



Figure 2-4 Functional Silos in Organization

- Silo environment is inefficient, inaccurate and expensive.
- Information is captured and **re-entered several times** and is **not available** in **real time**.
- Silo environments hamper enterprise decision making, because key information never makes it out of the different pockets of the organization **in time** for **decision maker**.



IS Architectures

- Rapid advances in computer and networking technologies and changing organizational dynamics, drive the emergence of new information system models.
- Web-based systems today use a distributed architecture which allows the sharing of applications and data resources between the client and the server computers.

IS Functionalization

- In addition to serving the different management levels, IS also supports major business functions, such as manufacturing, marketing, accounting, finance, and HR.
- Each functional area has different information needs and report requirements.
- Each functional area in an organization also has multiple levels of management, each requiring different levels of analysis and details of information.

Figure 2-5 Information Systems Architectures

- Three major types of information systems architecture have been commonly used in organizations:
Centralized,
Decentralized and
Distributed systems architecture.

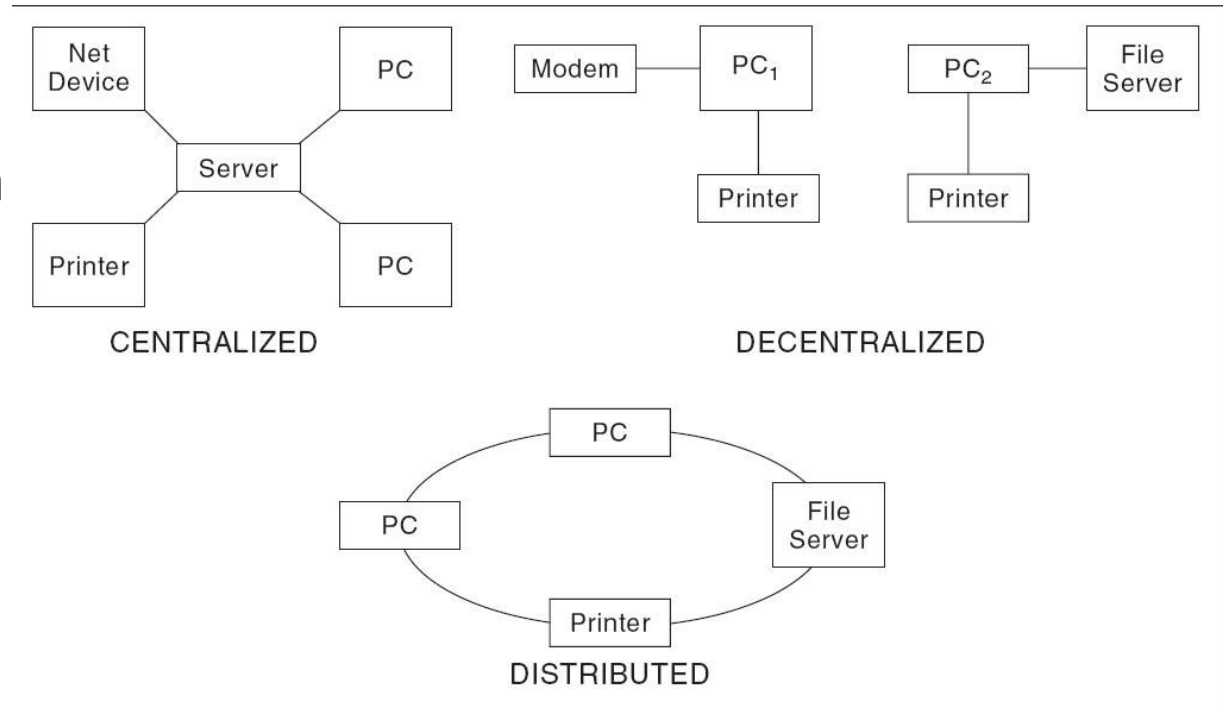
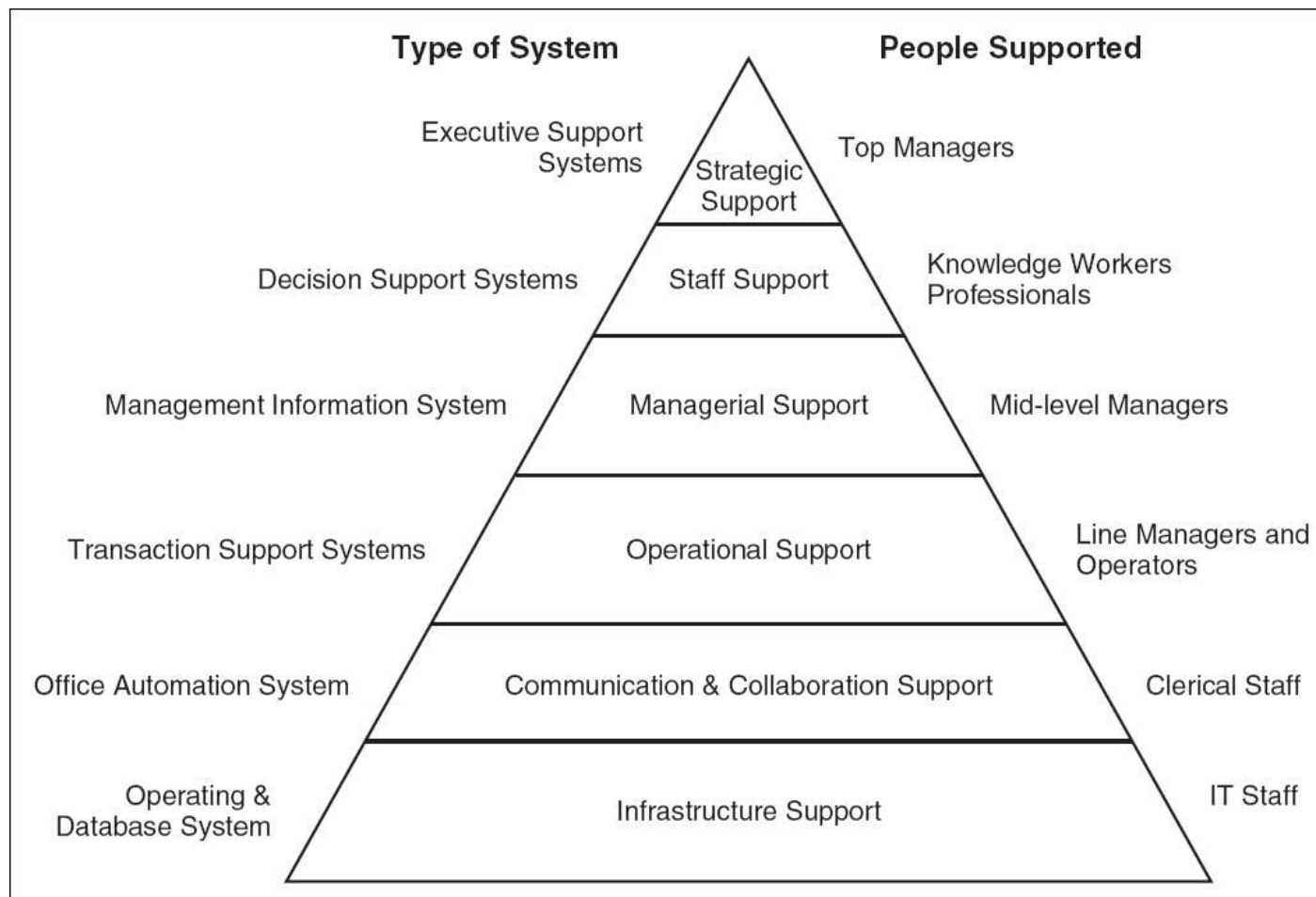


Figure 2-6 IS as Categorized by Functional and Hierarchical Models



Systems Integration

Logical

- Develop information systems that allow organizations to share data with all of its stakeholders based on need and authorization.
- Management needs to change organizational structures, processes, and employee roles and responsibilities.

Physical

- Provide seamless connectivity between heterogeneous systems.
- Business process reengineering involves changing the mindset of the employees in the organization, encouraging and enabling them to do their tasks in a new way.

Steps in Integrating Systems

Step 1	Resource categorization	Instituting IT support for an integrated systems environment is necessary to avoid support and maintenance problems with the integrated system.
Step 2	Compliance and standards	Develop a single sign-on policy because all employees and external partners will need access to an integrated system from anywhere, anytime.
Step 3	Legacy systems support	Develop a policy in support of older legacy applications.
Step 4	Middleware tools	Middleware tools are essential for integration in the short term if existing applications must be used by the organization.

Steps in Integrating Systems (Cont'd)

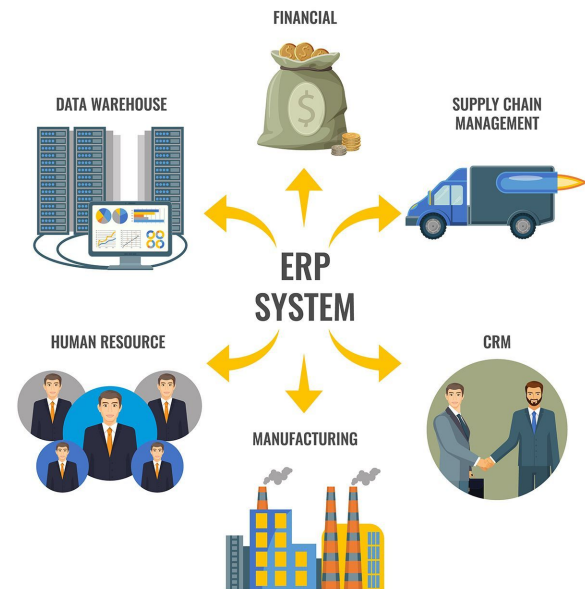
Step 5	Authentication and authorization policies	Single sign-on policy for application and data access because all employees and external partners will need access to an integrated system from anywhere, anytime.
Step 6	Centralized IT services and support	The IT staff needs to be able to support all applications and platforms with a centralized IT help desk support.
Step 7	Back-up, recovery, and security	A good back-up and recovery system is essential if there is a system failure or a major disaster.
Step 8	Hardware and software standardization	Develop organization standards and policy on acquisition of new hardware and software which is aligned with organization IT strategy.

Benefits and Limitations of Systems Integration

Benefits	Limitations
Increased Revenue and Growth	High Initial Set-up Costs
Leveling the Competitive Environment	Power and Interdepartmental Conflicts (due to the sharing of information)
Enhanced Information Visibility	Long-term and Intangible ROI (Usually several years)
Increased Standardization	Creativity Limitations (Restricts Creativity and Independence)

ERP and Systems Integration

- Enterprise Resource Planning (ERP) systems are integrated, multi-module application software packages designed to serve and support several business functions across an organization.
- ERP systems are typically commercial software packages that facilitate collection and integration of information related to various areas of an organization.



ERP's Role in Logical Integration

- ERP systems require organizations to focus on business process rather than on functions.
- ERP systems come with built-in processes for a wide variety of common business functions.



ERP's Role in Physical Integration

- Before installing the ERP system, an organization may have to upgrade or install middleware or get rid of their legacy system's hardware and software.
- Integration is also required at the Data level, Client level, and at the Application level.
- A good ERP implementation improves operational efficiency with better business processes that focuses on organizational goals rather than on individual departmental goals.

Implications for Management

- **Silos do not work.**
 - Most organizations lose out in the long-term when information is not shared in real time across the functional boundaries within the company.
- **System integration has many hidden benefits.**
 - Allows decision making to be cascaded to all departments
 - Allows employees at lower-levels to make better decisions while interacting with clients or partners.
- **System integration has many challenges.**
 - Replacing old hardware and software
 - Working with IT consultants
 - Human challenges, such as impact on IT staff, department heads losing control of data, and rumors of layoffs

Implications for Management (Cont'd)

- **Systems integration raises many new ethical issues.**
 - Possibility of some employees exploiting information for personal advantage and illegal access of information.
- Remedies can consist of:
 - Develop policies on ethical usage of information.
 - Install proper security software and hardware (like firewalls).
 - Allocate resources for training and education on accessing information.

Summary

- Functional silos categorize an organization's tasks and activities into groups to improve efficiency and responsibility of work in the organization.
- Silos can improve productivity, but they often lead employees to achieve departmental goals rather than overall organizational goals.
- IS over the years have been divided horizontally by functions and vertically by hierarchical levels.
- IS architecture has evolved from centralized mainframe architecture to personal computers with distributed or client–server architecture.

Summary (Cont'd)

- In order for systems integration to be successful, organizations have to focus both on the human or logical level and on the physical or systems level.
- ERP systems thus make the process of systems integration easier, but they are expensive and often require organizations to start from scratch.
- System integration involves the whole organization, requiring top-management support and resources for a long-term period. Management must be ready to face the human and ethical challenges in a systems integration project.